

SOME MIXTURES OF SINE-WAVE CIRCULAR DISTRIBUTIONS

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1. Introduction

Let $\mathcal{F} = \{F\}$ be a family of distributions and μ a measure on a Borel field of subsets of $\mathcal{F}$ with $\mu(\mathcal{F}) = 1$. Then $H(\cdot) = \int_{\mathcal{F}} F(\cdot) d\mu(F)$ is again a distribution function called a μ mixture of $\mathcal{F}$. In the case of an absolutely continuous distribution the mixture can be written as $\int f(x, \theta) g(\theta) d\theta$, where $g(\theta)$ is the probability density function of the parameter θ which indexes the family $\{f\}$ of probability density functions.

It seems that no attention has yet been paid to the problem of mixtures of circular distributions defined on the boundary of a circle. In this paper we prove a general result that any mixture of a sine-wave distribution is a sine-wave distribution. As an illustration, a mixture of a sine-wave distribution using the Von Mises circular distribution as distribution of the location parameter is obtained.

2. Some elements of the theory of circular distributions

Let Ω be the boundary of a unit circle. A circular distribution on Ω can be defined in the following way. Corresponding to a point on Ω , we can find a unique angle α , its vectorial angle restricted to a half-open interval I of length 2π and vice versa. Hence, the probability space Ω can be described as

$$(2.1) \quad \Omega = \{\alpha \mid \alpha \in (0, 2\pi]\}.$$

We use the random angle A (capital alpha) and denote its distribution function by $F(\alpha)$. For an absolutely continuous distribution, which we confine to in this paper, we denote the density function of the random angle A as $f(\alpha)$ which is the derivative of $F(\alpha)$.

2.1 Trigonometric Moments

Definition 2.1.1. The Trigonometric Moments of a circular distribution of the random angle A having the density function $f(\alpha)$ are the functions $\Phi_f(\nu)$ defined, for integer values ν , by

$$(2.2) \quad \Phi_f(\nu) = \int_0^{2\pi} \exp(i\nu\alpha) f(\alpha) d\alpha.$$

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It follows that $\Phi_f(0) = 1$ and $\Phi_f(-\nu)$ equals the complex conjugate $\overline{\Phi_f(\nu)}$ of $\Phi_f(\nu)$. Also $|\Phi_f(\nu)| \leq 1$ for all ν . For more properties of $\Phi_f(\nu)$, see Batschelet (1963) and Shohat and Tamarkin (1963). Hurwitz (1903) showed that the trigonometric moments determine a circular distribution uniquely.

2.2 Von Mises and Sine-wave distributions

For statistical applications Von Mises (1918) proposed a circular distribution having the density function

$$(2.3) f(\alpha) = \frac{\exp[k \cos(\alpha - \theta)]}{2\pi I_0(k)}, \quad \alpha, \theta \in (0, 2\pi], k \geq 0,$$

where $I_\nu(k)$, for any integer ν , denotes the ν -th Bessel function with purely imaginary argument. For this distribution the trigonometric moments are given by

$$\Phi_f(0) = 1 \text{ and } \Phi_f(\nu) = [I_\nu(k) / I_0(k)] \exp(i\nu\theta), \nu \neq 0.$$

Another well-known circular distribution, called the Sine-Wave distribution, has the density function

$$(2.4) f(\alpha) = (1/\pi) \left(\frac{1}{2} + \rho \cos(\alpha - \alpha_0) \right),$$

$$\alpha \in (0, 2\pi], \alpha_0 \in (0, 2\pi], 0 \leq \rho \leq \frac{1}{2}$$

where ρ and α_0 are the parameters; ρ is a parameter of concentration. Its trigonometric moments are given by

$$\Phi_f(0) = 1, \Phi_f(\nu) = \rho \exp(-i\nu\alpha_0), \nu = \pm 1$$

and

$$\Phi_f(\nu) = 0, \nu = \pm 2, \pm 3, \dots$$

The non-zero moments $\Phi_f(0)$, $\Phi_f(1)$ and $\Phi_f(-1)$ are about $\alpha = \alpha_0$, $\alpha_0 \neq 2\pi$. A linear representation of this density function is a full period of a sine-curve and hence the name.

3. Mixtures of sine-wave distribution

In this section, we show that any mixture of a sine-wave distribution is a sine-wave distribution. A relation between the trigonometric moments of a mixture of a circular distribution and the original distribution is mentioned. An illustration of the mixture of a sine-wave distribution is given.

Theorem 1.

Let A be a random variable having a sine-wave distribution, then any ρ mixture of this distribution is also a sine-wave distribution.

Proof:

The density of the mixture is given by

$$(3.1) \quad h(\alpha, \alpha_0) = \int_0^{1/2} (1/\pi) \left[\frac{1}{2} + \rho \cos(\alpha - \alpha_0) \right] g(\rho) d\rho$$

where $g(\rho)$ is the mixing density function. The above relation can be written as

$$(3.2) \quad h(\alpha, \alpha_0) = \frac{1}{2\pi} + \frac{\cos(\alpha - \alpha_0)}{\pi} \int_0^{1/2} \rho g(\rho) d\rho$$

$$= \frac{1}{\pi} \left(\frac{1}{2} + E(\rho) \cos(\alpha - \alpha_0) \right),$$

where $E(\rho)$ denotes the expected value of the distribution of ρ . It may be noted that ρ of the original distribution has been replaced by $E(\rho)$ in the mixture, and since $0 \leq \rho \leq \frac{1}{2}$, $0 < E(\rho) < \frac{1}{2}$. This shows that ρ mixture of a sine-wave distribution is also a sine-wave distribution.

Theorem 2.

Let A be a random variable having a sine-wave distribution, then any α_0 mixture of this distribution is also a sine-wave distribution.

Proof:

The density of the mixture is given by

$$(3.3) \quad h(\alpha, \rho) = \int_0^{2\pi} \frac{1}{\pi} \left[\frac{1}{2} + \rho \cos(\alpha - \alpha_0) \right] g(\alpha_0) d\alpha_0,$$

where $g(\alpha_0)$ is the mixing density function. The above relation can be written as

$$(3.4) \quad h(\alpha, \rho) = \frac{1}{2\pi} + \frac{\rho}{\pi} \operatorname{Re} \int_0^{2\pi} \exp(i(\alpha - \alpha_0)) g(\alpha_0) d\alpha_0$$

$$= \frac{1}{2\pi} + \frac{\rho}{\pi} \operatorname{Re} [\exp(i\alpha) \overline{\mu_1(g)}]$$

where $\overline{\mu_1(g)}$ is the complex conjugate of the first trigonometric moment of the

distribution of α_0 . Writing $\overline{\mu_1(g)} = u_1 - iv_1$, we obtain

$$\begin{aligned}
 h(\alpha, \rho) &= \frac{1}{2\pi} + \frac{\rho}{\pi} \Re \ell [(\cos \alpha + i \sin \alpha)(u_1 - iv_1)] \\
 &= \frac{1}{2\pi} + \frac{\rho}{\pi} [u_1 \cos \alpha + v_1 \sin \alpha] \\
 &= \frac{1}{2\pi} + \frac{\rho}{\pi} \left[\frac{u_1}{\sqrt{u_1^2 + v_1^2}} \cos \alpha + \frac{v_1}{\sqrt{u_1^2 + v_1^2}} \sin \alpha \right] \sqrt{u_1^2 + v_1^2} \\
 (3.5) \quad &= \frac{1}{2\pi} + \frac{\rho}{\pi} \sqrt{u_1^2 + v_1^2} \cos(\alpha - \beta).
 \end{aligned}$$

where β is uniquely determined modulo 2π by the relations

$$\cos \beta = \frac{u_1}{\sqrt{u_1^2 + v_1^2}}, \quad \sin \beta = \frac{v_1}{\sqrt{u_1^2 + v_1^2}} \quad \text{Thus}$$

$$(3.6) \quad h(\alpha, \rho) = \frac{1}{\pi} \left[\frac{1}{2} + \rho \sqrt{u_1^2 + v_1^2} \cos(\alpha - \beta) \right],$$

which is again the density of a sine-wave distribution.

Remark:

Note that the α_0 mixture depends only on the first trigonometric moment of the mixing density function.

Theorem 3.

Let A be a random variable having a sine-wave distribution with parameters α_0 and ρ . Then any α_0 - ρ mixture of this distribution is also a sine-wave distribution.

Proof:

Let $g(\alpha_0, \rho)$ denote the joint mixing probability density function of the parameters α_0 and ρ . Then the marginal probability density function of α_0 is given by

$$(3.7) \quad g_1(\alpha_0) = \int_0^{1/2} g(\alpha_0, \rho) d\rho.$$

The density of the mixture is

$$\begin{aligned} h(\alpha) &= \int_0^{2\pi} \int_0^{1/2} \frac{1}{\pi} \left(\frac{1}{2} + \rho \cos(\alpha - \alpha_0) \right) g(\alpha_0, \rho) d\rho d\alpha_0 \\ &= \frac{1}{2\pi} + \frac{1}{\pi} \int_0^{2\pi} \cos(\alpha - \alpha_0) E(\rho | \alpha_0) g_1(\alpha_0) d\alpha_0 \\ (3.8) \quad &= \frac{1}{2\pi} + \frac{1}{\pi} E(\rho) \int_0^{2\pi} \cos(\alpha - \alpha_0) \frac{E(\rho | \alpha_0) g_1(\alpha_0)}{E(\rho)} d\alpha_0 \end{aligned}$$

where $E(\rho | \alpha_0)$ and $E(\rho)$ are, respectively, conditional and absolute expectations of ρ . This is an α_0 mixture of a sine-wave distribution, with parameters $E(\rho)$ and α_0 , the mixing density being $E(\rho | \alpha_0) g_1(\alpha_0) / E(\rho)$. Thus by theorem 2, the result is established.

The following lemma, proof of which is straightforward, establishes a relationship between the trigonometric moments of a mixture of a circular distribution and the original distribution.

Lemma 1:

Let A be a random variable having a circular distribution with p.d.f. $f(\alpha, \theta)$, then the ν -th trigonometric moment $\Phi_h(\nu)$ of its θ mixture is given as

$$(3.9) \quad \Phi_h(\nu) = \int \Phi_f(\nu, \theta) g(\theta) d\theta,$$

where $\Phi_f(\nu, \theta)$ is the ν -th trigonometric moment of the distribution of A and $g(\theta)$ is the mixing p.d.f. of θ .

This lemma is very useful in obtaining the trigonometric moments of the mixture by using the trigonometric moments of the original distribution, because in many instances, the mixture itself has a complicated density function.

We now give an illustration of a mixture of a sine-wave distribution.

3.1 Mixing with Von Mises Distribution

We write the density function of sine-wave distribution as

$$(3.10) \quad f(\alpha, \rho, \alpha_0) = (1/\pi) \left(\frac{1}{2} + \rho \cos(\alpha - \alpha_0) \right), \quad \alpha, \alpha_0 \in (0, 2\pi], 0 \leq \rho \leq \frac{1}{2},$$

where ρ is fixed and α_0 has the Von Mises distribution having the density function

$$(3.11) \quad g(\alpha_0) = \frac{\exp[k \cos(\alpha_0 - \theta)]}{2\pi I_0(k)}, \quad \alpha_0, \theta \in (0, 2\pi], k > 0.$$

Here $\mu_1(g) = \frac{I_1(k)}{I_0(k)} \exp(i\theta)$ and, therefore, $\overline{\mu_1(g)} = \frac{I_1(k)}{I_0(k)} \exp(-i\theta) = u_1 - iv_1$.

Thus u_1 and v_1 in the proof of theorem 2 are given by $u_1 = \frac{I_1(k)}{I_0(k)} \cos \theta$ and $v_1 = \frac{I_1(k)}{I_0(k)} \sin \theta$. Therefore, $\sqrt{u_1^2 + v_1^2} = \frac{I_1(k)}{I_0(k)}$ and $\beta \equiv \theta$ and

$$(3.12) \quad h(\alpha, \rho) = \frac{1}{\pi} \left[\frac{1}{2} + \rho \frac{I_1(k)}{I_0(k)} \cos(\alpha - \theta) \right].$$

This shows that the mixture is a sine-wave distribution with parameters

$$\theta \text{ and } \rho \frac{I_1(k)}{I_0(k)}, \quad 0 \leq \rho \frac{I_1(k)}{I_0(k)} \leq \frac{1}{2}, \quad \theta \in (0, 2\pi].$$

Lemma 1 and the property of uniqueness of trigonometric moments also enables us to reach the above conclusion.

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